Maharashtra Education Society's

Institute of Management and Career Courses (IMCC), Pune

AUTONOMOUS

IT13: ADBMS
Chapter 2:SQL
Practical Assignment 1 Solution Template

Write the SQL query and paste a screenshot of the output below the question.

1. **Create a database and two tables in it.**
2. **Insert at least 5 rows into the table.**
3. **Display all the information in the Employee table.**
4. **Add columns 'Age' and 'Salary' to the Employee table; apply check constraints for (Age<=50 and Salary>=10000).**
5. **Add a column 'Age' to the Investor table.**
6. **Change the data type of inv_name from varchar to text in the Investor table**
7. **Change the name of the column/field Emp_id to E_id.**
8. **Update the city of Emp_id-4 to the current city, Nagpur.**
9. **Delete one record from the Investor table.**
10. **Drop the column 'Age' from the Investor table.**
11. **Drop the CHECK constraint on the 'Age' column from the Employee table.**
12. **Create foreign key constraints between the Employee and Investor tables**
13. **Display any 4 columns from the Investor table.**
14. **Rename a table.**
15. **Delete all the records using the TRUNCATE command.**
16. **Drop both tables from the database.**

1.

```
mysql> USE CollegeDB;
Database changed
mysql>
mysql> CREATE TABLE Employee (
    ->     Emp_id INT PRIMARY KEY,
    ->     Emp_name VARCHAR(50),
    ->     City VARCHAR(50),
    ->     Department VARCHAR(50)
    -> );

mysql>
mysql> CREATE TABLE Investor (
    ->     Inv_id INT PRIMARY KEY,
    ->     inv_name VARCHAR(50),
    ->     Investment DECIMAL(10,2),
    ->     Emp_id INT
    -> );

mysql> SHOW DATABASES;
```

```
+--------------------+
| Database           |
+--------------------+
| adbms_practice     |
| collegedb          |
| companydb          |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
```

2.

```
mysql> INSERT INTO Employee VALUES
    -> (1, 'Amit', 'Pune', 'IT'),
    -> (2, 'Priya', 'Mumbai', 'HR'),
    -> (3, 'Rahul', 'Nashik', 'Finance'),
    -> (4, 'Sneha', 'Pune', 'IT'),
    -> (5, 'Rohan', 'Nagpur', 'Sales');
```

```
mysql> INSERT INTO Investor VALUES
    -> (101, 'Rajesh', 50000.00, 1),
    -> (102, 'Neha', 75000.00, 2),
    -> (103, 'Suresh', 60000.00, 3),
    -> (104, 'Pooja', 90000.00, 4),
    -> (105, 'Kiran', 40000.00, 5);
```

3.

```
mysql> SELECT * FROM Employee;
```

Emp_id	Emp_name	City	Department
1	Amit	Pune	IT
2	Priya	Mumbai	HR
3	Rahul	Nashik	Finance
4	Sneha	Pune	IT
5	Rohan	Nagpur	Sales

4.

```
mysql> ALTER TABLE Employee
    -> ADD Age INT CONSTRAINT chk_age CHECK (Age <= 50),
    -> ADD Salary DECIMAL(10,2) CONSTRAINT chk_salary CHECK (Salary >= 10000);

mysql> DESCRIBE Employee;
```

Field	Type	Null	Key	Default	Extra
Emp_id	int	NO	PRI	NULL	
Emp_name	varchar(50)	YES		NULL	
City	varchar(50)	YES		NULL	
Department	varchar(50)	YES		NULL	
Age	int	YES		NULL	
Salary	decimal(10,2)	YES		NULL	

5.

```
mysql> ALTER TABLE Investor
    -> ADD Age INT;
```

```
mysql> DESCRIBE Investor;
```

Field	Type	Null	Key	Default	Extra
Inv_id	int	NO	PRI	NULL	
inv_name	varchar(50)	YES		NULL	
Investment	decimal(10,2)	YES		NULL	
Emp_id	int	YES		NULL	
Age	int	YES		NULL	

6.

```
mysql> ALTER TABLE Investor
    -> MODIFY inv_name TEXT;
```

```
mysql> DESCRIBE Investor;
```

Field	Type	Null	Key	Default	Extra
Inv_id	int	NO	PRI	NULL	
inv_name	text	YES		NULL	
Investment	decimal(10,2)	YES		NULL	
Emp_id	int	YES		NULL	
Age	int	YES		NULL	

7.

```
mysql> ALTER TABLE Employee
   -> RENAME COLUMN Emp_id TO E_id;

mysql> DESCRIBE Employee;
```

```
+------------+---------------+------+-----+---------+-------+
| Field      | Type          | Null | Key | Default | Extra |
+------------+---------------+------+-----+---------+-------+
| E_id       | int           | NO   | PRI | NULL    |       |
| Emp_name   | varchar(50)   | YES  |     | NULL    |       |
| City       | varchar(50)   | YES  |     | NULL    |       |
| Department | varchar(50)   | YES  |     | NULL    |       |
| Age        | int           | YES  |     | NULL    |       |
| Salary     | decimal(10,2) | YES  |     | NULL    |       |
+------------+---------------+------+-----+---------+-------+
```

8.

```
mysql> UPDATE Employee
   -> SET City = 'Nagpur'
   -> WHERE E_id = 4;

mysql> SELECT * FROM Employee
   -> WHERE E_id = 4;
```

```
+------+----------+--------+------------+------+--------+
| E_id | Emp_name | City   | Department | Age  | Salary |
+------+----------+--------+------------+------+--------+
|    4 | Sneha    | Nagpur | IT         | NULL |   NULL |
+------+----------+--------+------------+------+--------+
```

9.

```
mysql> DELETE FROM Investor
   -> WHERE Inv_id = 105;
```

```
mysql> SELECT * FROM Investor;
```

Inv_id	inv_name	Investment	Emp_id	Age
101	Rajesh	50000.00	1	NULL
102	Neha	75000.00	2	NULL
103	Suresh	60000.00	3	NULL
104	Pooja	90000.00	4	NULL

10.

```
mysql> ALTER TABLE Investor
   -> DROP COLUMN Age;
```

```
mysql> DESCRIBE Investor;
```

Field	Type	Null	Key	Default	Extra
Inv_id	int	NO	PRI	NULL	
inv_name	text	YES		NULL	
Investment	decimal(10,2)	YES		NULL	
Emp_id	int	YES		NULL	

11.

```
mysql> ALTER TABLE Employee
   -> DROP CHECK chk_age;
```

mysql> SHOW CREATE TABLE Employee;

```
| Employee | CREATE TABLE `employee` (
  `E_id` int NOT NULL,
  `Emp_name` varchar(50) DEFAULT NULL,
  `City` varchar(50) DEFAULT NULL,
  `Department` varchar(50) DEFAULT NULL,
  `Age` int DEFAULT NULL,
  `Salary` decimal(10,2) DEFAULT NULL,
  PRIMARY KEY (`E_id`),
  CONSTRAINT `chk_salary` CHECK ((`Salary` >= 10000))
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
```

12.

```
mysql> ALTER TABLE Investor
   -> ADD CONSTRAINT FK_Investor_Employee
   -> FOREIGN KEY (Emp_id)
   -> REFERENCES Employee(E_id);
```

mysql> SHOW CREATE TABLE Investor;

```
| Investor | CREATE TABLE `investor` (
  `Inv_id` int NOT NULL,
  `inv_name` text,
  `Investment` decimal(10,2) DEFAULT NULL,
  `Emp_id` int DEFAULT NULL,
  PRIMARY KEY (`Inv_id`),
  KEY `FK_Investor_Employee` (`Emp_id`),
  CONSTRAINT `FK_Investor_Employee` FOREIGN KEY (`Emp_id`) REFERENCES `employee` (`E_id
`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
```

13.

```
mysql> SELECT Inv_id, inv_name, Investment, Emp_id
    -> FROM Investor;
```

```
+--------+----------+------------+--------+
| Inv_id | inv_name | Investment | Emp_id |
+--------+----------+------------+--------+
|    101 | Rajesh   |   50000.00 |      1 |
|    102 | Neha     |   75000.00 |      2 |
|    103 | Suresh   |   60000.00 |      3 |
|    104 | Pooja    |   90000.00 |      4 |
+--------+----------+------------+--------+
```

14.

```
mysql> RENAME TABLE Investor TO InvestorDetails;

mysql> SHOW TABLES;
```

```
+---------------------+
| Tables_in_collegedb |
+---------------------+
| employee            |
| investordetails     |
+---------------------+
```

15.

```
mysql> SET FOREIGN_KEY_CHECKS = 0;

mysql> TRUNCATE TABLE InvestorDetails;
mysql> TRUNCATE TABLE Employee;

mysql> SET FOREIGN_KEY_CHECKS = 1;

mysql> SELECT * FROM Employee;
```

```
Empty set (0.02 sec)
```

```
mysql> SELECT * FROM InvestorDetails;
```

```
Empty set (0.00 sec)
```

16.

mysql> DROP TABLE InvestorDetails;

mysql> DROP TABLE Employee;

mysql> SHOW TABLES;

```
Empty set (0.00 sec)
```